

Exercise C1

- Q1. Let $\frac{3\pi}{4} < \alpha < \pi$, then $\sqrt{2 \cot \alpha + \frac{1}{\sin^2 \alpha}}$ is equal to :
- (A) $\frac{\sin \alpha - \cos \alpha}{\sin \alpha}$ (B) $\frac{\sin \alpha + \cos \alpha}{\sin \alpha}$ (C) $\frac{-\sin \alpha + \cos \alpha}{\sin \alpha}$ (D) $\frac{-\sin \alpha - \cos \alpha}{\sin \alpha}$

Ans. D

Sol. -

- Q2. If $\cos^2 A + \cos^2 B + \cos^2 C = 1$, then triangle ABC is :
- (A) right angled (B) equilateral (C) isosceles (D) None of these

Ans. A

Sol. -

- Q3. If $\frac{\tan 3A}{\tan A} = k$, then $\frac{\sin 3A}{\sin A}$ is equal to :
- (A) $\frac{2k}{k-1}$ (B) $\frac{2k}{k+1}$ (C) $\frac{3k}{k+1}$ (D) None of these

Ans. A

Sol. -

- Q4. $3(\sin x - \cos x)^4 + 6(\sin x + \cos x)^2 + 4(\sin^6 x + \cos^6 x) =$
- (A) 11 (B) 12 (C) 13 (D) 14

Ans. C

Sol. -

- Q5. $\sec^2 \theta = \frac{4xy}{(x+y)^2}$ is true if and only if :
- (A) $x + y \neq 0$ (B) $x = y, x \neq 0$ (C) $x = y$ (D) $x \neq 0, y \neq 0$

Ans. B

Sol. $\operatorname{cosec}^2 \theta \geq 1$ and

$$\frac{4xy}{(x+y)^2} \leq 1$$

So LHS = 1 and RHS = 1

$$\frac{4xy}{(x+y)^2} = 1$$

$$\Rightarrow (x-y)^2 = 0$$

$$\Rightarrow x = y$$

but $x \neq 0$ ($\operatorname{cosec}^2 \theta \neq 0$)

Q6. If $\alpha + \beta = \frac{\pi}{2}$ and $\beta + \gamma = \alpha$, then $\tan \alpha$ equals :

- (A) $2 (\tan \beta + \tan \gamma)$ (B) $\tan \beta + \tan \gamma$ (C) $\tan \beta + 2 \tan \gamma$ (D) $2 \tan \beta + \tan \gamma$

Ans. C

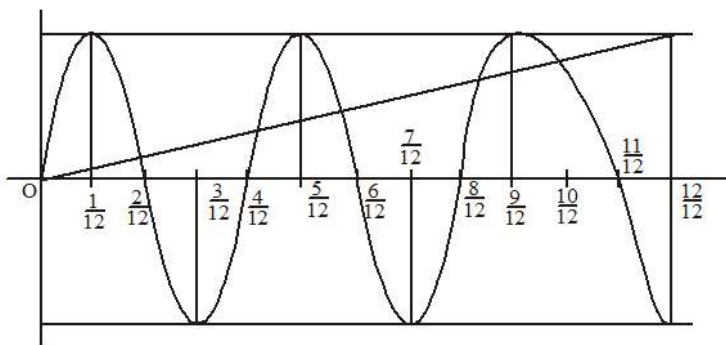
Sol. -

Q7. Number of real solutions to the equation $\sin(6\pi x) = x$, is

- (A) 13 (B) 11 (C) 9 (D) 7

Ans. B

Sol. Draw graphs of $y = x$ and $y = \sin 6\pi x$ and the intersect.



Q8. The value of x satisfying the equation, $x = \sqrt{2 + \sqrt{2 - \sqrt{2 + x}}}$ is

- (A) $2 \cos 10^\circ$ (B) $2 \cos 20^\circ$ (C) $2 \cos 40^\circ$ (D) $2 \cos 80^\circ$

Ans. C

Sol. Note that $x \in [-2, 2]$ for x for x to be real

Let $x = 2 \cos \theta$ where $\theta \in [0, \pi]$

$$x = \sqrt{2 + \sqrt{2 - \sqrt{2 + 2 \cos \theta}}}$$

$$2 \cos \theta = \sqrt{2 + \sqrt{2 - 2 \cos \frac{\theta}{2}}} = \sqrt{2 + \sqrt{2 \left(1 - \cos \frac{\theta}{2}\right)}} = \sqrt{2 + 2 \sin \frac{\theta}{4}} = \sqrt{2 + 2 \cos \left(\frac{\pi}{2} - \frac{\theta}{4}\right)}$$

$$= \sqrt{2 \left(1 + \cos \left(\frac{\pi}{2} - \frac{\theta}{4}\right)\right)} = \sqrt{4 \cos^2 \left(\frac{\pi}{4} - \frac{\theta}{8}\right)}$$

$$2 \cos \theta = 2 \cos \left(\frac{\pi}{4} - \frac{\theta}{8}\right); \quad \therefore \theta = \frac{\pi}{4} - \frac{\theta}{8} \Rightarrow \frac{9\theta}{8} \Rightarrow \theta = \frac{2\pi}{9}$$

Hence $x = 2 \cos \frac{2\pi}{9} = 2 \cos 40^\circ \Rightarrow$ (C) Ans.

Q9. The value of, $(\tan 18^\circ)(\sin 36^\circ)(\cos 54^\circ)(\tan 72^\circ)(\tan 108^\circ)(\cos 126^\circ)(\sin 144^\circ)(\tan 162^\circ)(\cos 180^\circ)$ is $k \sin^2 18^\circ$ then k has the value equal to

- (A) $\frac{5}{16}$ (B) $\frac{5}{4}$ (C) $\frac{1}{4}$ (D) $\frac{1}{16}$

Ans. B

Sol. $(\tan 18^\circ)(\tan 72^\circ)(\tan 108^\circ)(\tan 162^\circ) = 1$ hence we are left with

$$(-)(\sin 36^\circ)(\cos 54^\circ)(\cos 126^\circ)(\sin 144^\circ)$$

$$(-)(\sin 36^\circ)(\sin 36^\circ)(-36^\circ)(\sin 36^\circ) = \sin^4 36^\circ$$

$$= \left(\frac{1 - \cos 72^\circ}{2} \right) = \frac{(1 - \sin 18^\circ)^2}{4}$$

$$= \frac{1}{4} \left(1 - \frac{\sqrt{5} - 1}{4} \right)^2 = \frac{1}{4} \left(\frac{5 - \sqrt{5}}{4} \right)^2 = \frac{5}{4} \left(\frac{\sqrt{5} - 1}{4} \right) = \frac{5}{4} \sin^2 18^\circ \Rightarrow k = \frac{5}{4} \text{ Ans.}$$

Q10. If $x \sec \alpha + y \tan \alpha = x \sec \beta + y \tan \beta = a$, then $\sec \alpha \cdot \sec \beta =$

- (A) $\frac{a^2 + y^2}{x^2 + y^2}$ (B) $\frac{a^2 + y^2}{x^2 - y^2}$ (C) $\frac{x^2 + y^2}{a^2 + y^2}$ (D) $\frac{x^2 - y^2}{a^2 - y^2}$

Ans. B

Sol. α and β satisfy the equation

$$x \sec \theta + y \tan \theta = a$$

$$\text{or } (\sec \theta - a^2) = y^2 \tan^2 = y^2 (x \sec^2 - 1)$$

$$\sec^2 \theta (x^2 - y^2) - ax \sec \theta + a^2 + y^2 = 0$$

This is a quadratic in $\sec \theta$, whose roots are $\sec \alpha$ and $\sec \beta$

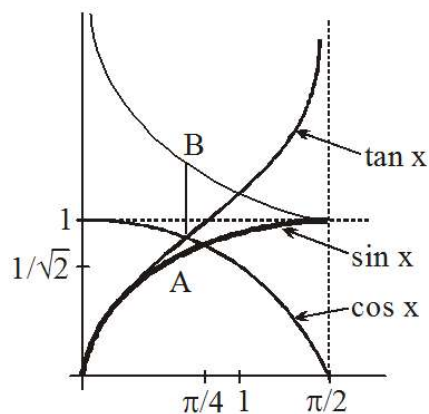
$$\sec \alpha \cdot \sec \beta = \frac{a^2 + y^2}{x^2 - y^2}$$

Q11. The graphs of $y = \sin x$, $y = \cos x$, $y = \tan x$ and $y = \operatorname{cosec} x$ are drawn on the same axes from 0 to $\pi/2$. A vertical line is drawn through the point where the graphs of $y = \cos x$ and $y = \tan x$ cross, intersecting the other two graphs at points A and B. The length of the line segment AB is:

- (A) 1 (B) $\frac{\sqrt{5} - 1}{2}$ (C) $\sqrt{2}$ (D) $\frac{\sqrt{5} + 1}{2}$

Ans. A

Sol.



Given $\tan x = \cos x$

or $\sin x = \cos^2 x = 1 - \sin^2 x$ (1)

now, $\operatorname{cosec} x - \sin x = \frac{1 - \sin^2 x}{\sin x} = 1$ from (1)

Q12. The value of $4\cos \frac{\pi}{10} - 3 \sec \frac{\pi}{10} - 2\tan \frac{\pi}{10}$ is equal to

- (A) 1 (B) $\sqrt{5} - 1$ (C) $\sqrt{5} + 1$ (D) Zero

Ans. D

Sol. $4 \cos 18^\circ - \frac{3}{\cos 18^\circ} - 2 \tan 18^\circ$

$$\frac{4 \cos^2 18^\circ - 3 - 2 \sin 18^\circ}{\cos 18^\circ} = \frac{2(1 + \cos 36^\circ) - 2 \sin 18^\circ - 3}{\cos 18^\circ}$$

$$\Rightarrow \frac{2(1 + \cos 36^\circ - \sin 18^\circ) - 3}{\cos 18^\circ} = \frac{2\left(1 + \frac{1}{2}\right) - 3}{\cos 18^\circ} = 0$$

Q13. The value of $\cot x + \cot (60^\circ + x) + \cot (120^\circ + x)$ is equal to :

- (A) $\cot 3x$ (B) $\tan 3x$ (C) $3 \tan 3x$ (D) $\frac{3 - 9 \tan^2 x}{3 \tan x - \tan^3 x}$

Ans. D

Sol. $\cot x + \frac{\cos (60 + x)}{\sin (x - 60)} + \frac{\cos (x - 60)}{\sin (x - 60)}$

Note : $\cot \theta + \cot (60 + \theta) + \cot (180 - 60 + \theta)$

$$= \frac{\cos x}{\sin x} + \frac{\sin (2x)}{\sin (x + 60) \sin (x - 60)}$$

$$\cot \theta + \cot (60 + \theta) - \cot (60 - \theta) = 3 \cot 3$$

$$= \frac{\cos x}{\sin x} + \frac{8 \sin x \cos x}{4 \sin^2 x - 3} = \frac{4 \sin^2 x \cos x - 3 \cos x + 8 \sin^2 x \cos x}{4 \sin^3 x - 3 \sin x}$$

$$= \frac{3[3 \cos x - 4 \cos^3 x]}{\sin^3 x} = 3 \cot 3x \Rightarrow \frac{3[1 - 3 \tan^2 x]}{3 \tan x - \tan^3 x} \text{ Ans.}$$

Q14. The minimum value of the function $f(x) =$

$$\frac{\sin x}{\sqrt{1 - \cos^2 x}} + \frac{\cos x}{\sqrt{1 - \sin^2 x}} + \frac{\tan x}{\sqrt{\sec^2 - 1}} + \frac{\cot x}{\sqrt{\operatorname{cosec}^2 - 1}} \text{ as } x \text{ varies over all numbers in the largest possible domain of } f(x) \text{ is}$$

- (A) 4 (B) -2 (C) 0 (D) 2

Ans. B

Sol. If $x \in 1^{\text{st}}$ quadrant then $y = 1 + 1 + 1 + 1 = 4$

If $x \in 2^{\text{nd}}$ quadrant then $y = 1 - 1 - 1 - 1 = -2$

If $x \in 3^{\text{rd}}$ quadrant then $y = -1 - 1 + 1 + 1 = 0$

If $x \in 4^{\text{th}}$ quadrant then $y = -1 - 1 + 1 - 1 = -2$

$$\Rightarrow y_{\min} = -2$$

Q15. If $a > b > 0$, then the minimum value of $a \sec \theta - b \tan \theta \forall \theta \in \left(0, \frac{\pi}{2}\right)$

- (A) $\sqrt{a^2 - b^2}$ (B) $a - b$ (C) $\sqrt{a - b}$ (D) $(a^2 - b^2)^{3/2}$

Ans. A

Sol. Put $y + b \tan \theta = a \sec \theta$. Square and make a quadratic in $\tan \theta$. Put $D \leq 0$.

Note that min occurs when $\sin \theta = b/a$.

Q16. If $\sin(x - y) = \cos(x + y) = \frac{1}{2}$, then the values of 'x' and 'y' lying between 0° and 180° are given by

- (A) $x = 45^\circ, y = 15^\circ$ (B) $x = 45^\circ, y = 135^\circ$ (C) $x = 165^\circ, y = 15^\circ$
(D) $x = 165^\circ, y = 135^\circ$

Ans. A, D

Sol. -

Q17. For a positive integer 'n', let :

$$f_n(\theta) = \left(\tan \frac{\theta}{2}\right)(1 + \sec \theta)(1 + \sec 2\theta)(1 + \sec 4\theta) \dots (1 + \sec 2n\theta). \text{ Then :}$$

- (A) $f_2\left(\frac{\pi}{16}\right) = 1$ (B) $f_3\left(\frac{\pi}{32}\right) = 1$ (C) $f_4\left(\frac{\pi}{64}\right) = 1$ (D) $f_5\left(\frac{\pi}{128}\right) = 1$

Ans. A, B, C, D

Sol. -

Q18. If S is the solution set of all values of x for which $2 \cos t = \frac{3x+1}{x-1}$, then :

- (A) Number of integral values in S is 5 (B) Number of integral values in S is 4
 (C) Sum of all integral values in S is -6. (D) Sum of all integral values in S is -5.

Ans. B, C

Sol. $-2 \leq 2 \cos t \leq 2$

$$-2 \leq \frac{3x+1}{x-1} \leq 2$$

$$x \neq 1$$

Q19. Let $f_n(\theta) = 2(\sin^2\theta + \sin 2\theta \sin 4\theta + \sin 3\theta \sin 9\theta + \dots + \sin n\theta \sin n^2\theta)$ then which of the following hold(s) good :

- (A) $f_3\left(\frac{\pi}{6}\right) = 0$ (B) $f_4\left(\frac{\pi}{3}\right) = 2$ (C) $f_8\left(\frac{\pi}{2}\right) = 0$ (D) $f_{11}\left(\frac{\pi}{4}\right) = 2$

Ans. A, C, D

Sol.

$$f_n(\theta) = \sum_{r=1}^n 2 \sin r\theta \sin r^2\theta$$

$$= \sum_{r=1}^n [\cos(r^2 - r)\theta - \cos(r^2 + r)\theta]$$

$$= \cos 0^\circ - \cos(n^2 + n)\theta = 1 - \cos(n^2 + n)\theta$$

$$\therefore f_3\left(\frac{\pi}{6}\right) = 1 - \cos(9 + 3)\frac{\pi}{6} = 1 - \cos 2\pi = 0$$

$$f_8\left(\frac{\pi}{2}\right) = 1 - \cos(64 + 8)\frac{\pi}{2} = 1 - \cos 36\pi = 0$$

$$f_{11}\left(\frac{\pi}{4}\right) = 1 - \cos(121 + 11)\frac{\pi}{4} = 1 - \cos 33\pi = 2$$

$$f_4\left(\frac{\pi}{3}\right) = 1 - \cos\left(\frac{19\pi}{3}\right) = 1 - \cos \frac{\pi}{3} = \frac{1}{2}$$